



Problem solving with arithmetic procedures involving fractions

- 1 A small, rectangular plot of dirt has length $\frac{3}{4}$ metres by $\frac{7}{10}$ metres. Work out the area of the plot of dirt in m^2 . Tick **1** correct answer

☐ $\frac{30}{28} \text{ m}^2$

☐ $\frac{14}{11} \text{ m}^2$

☐ $\frac{11}{14} \text{ m}^2$

☒ $\frac{21}{40} \text{ m}^2$

☐ $\frac{15}{14} \text{ m}^2$

- 2 Andeep wants to put an edging around a small, rectangular plot of dirt. It has length $\frac{3}{4}$ m by $\frac{7}{10}$ m. Work out the total length of the edging, giving your answer in cm. Tick **1** correct answer

☐ 1.45 cm

☐ 145 cm

☒ 290 cm

☐ 29 cm

☐ 300 cm



- 3 A plot of land measures $5\frac{4}{5}$ m by $2\frac{9}{10}$ m. It will cost £20 per m^2 to grass this plot of land.

Who has correctly worked out the cost? Tick **1** correct answer

- ☐ Andeep says "Rounding, the area is $6 \times 3 = 18$ so $18 \times 20 = \text{£ } 360$ "
- ☐ Sofia says " 5.8×2.9 so rounding gives $6 \times 3 = 18$. Andeep is right."
- ☒ Jun says " $\frac{29}{5} \times \frac{29}{10} = 16\frac{41}{50}$, $17 \times 20 = \text{£ } 340$ "

- 4 A square picture frame is made from wood with length $\frac{4}{5}$ m. Laura wants to put a metal trim around the picture. The trim cost £1.20 per 10 cm. How much will it cost to trim the frame? Tick **1** correct answer

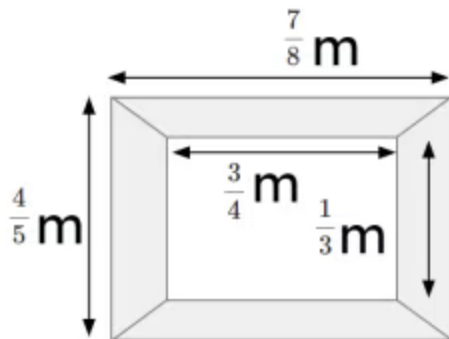
- ☐ £45.80
- ☐ £39.50
- ☐ £23.80
- ☒ £38.40
- ☐ £40

- 5 A piece of farmland has an area of $\frac{3}{5} \times 1\frac{2}{3} \text{ km}^2$. One sheep needs 2500 m^2 of land for grazing. How many sheep can live on the farmland? Fill in the blank

400



- 6 Aisha is restoring some old picture frames. She wants to cover the frame in gold leaf. Gold leaf costs £1 per cm^2 . How much will it cost Aisha to gold leaf the picture frame? Fill in the blank



45

